High-Level Design – Application Settings

TaskFlow – Intelligent Task Allocation & Tracking System

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Author: Jojo George (jojogeorge3344@gmail.com)
Based on: BRD-Application-Settings v1.0, FRD-Application-Settings v1.0
Module: Settings · **Sub-module:** 06.2 Application Settings · **Code:** APP
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related documents:

Document	Location
Overall HLD (stack, deployment, cross-cutting design)	`../../00-Overall/HLD-TaskFlow-Overall.md`
BRD – Application Settings	`../../../BRD/06-Settings/06.2-Application-Settings/BRD-Application-Settings.md`
FRD – Application Settings	`../../../FRD/06-Settings/06.2-Application-Settings/FRD-Application-Settings.md`
LLD – Application Settings	`../../../LLD/06-Settings/06.2-Application-Settings/LLD-Application-Settings.md`
HLD – Reminder Settings (shares the settings table)	`../06.1-Reminder-Settings/HLD-Reminder-Settings.md`
HLD – Notification (in-app switch consumer)	`../../04-Communication/04.2-Notification/HLD-Notification.md`
TDD	`TF - Documents/TF - TDD/TDD-TaskFlow.md` §7.5 (date, time, timezone), §8.9 (Settings)

1. Introduction

1.1 Purpose

This HLD describes the architecture of Application Settings: the organization-wide brand name, application timezone, default task priority and in-app notifications switch; the components that store

and serve them; and how the rest of TaskFlow consumes them – above all the single application clock used for every "today", "due today" and "overdue" decision. Class- and column-level detail is in the LLD.

1.2 Scope

In scope: the Application Settings screen (`/settings/application`), its API (`/api/v1/settings/application`, read and save), the settings service, the application clock (`AppClock`), storage in the shared key/value table, start-up defaults, and the read paths used by the shell, Create Task and the notification publisher.

Out of scope: reminder intervals and Mail delivery (Reminder Settings), notification creation and display (Notification), change history of settings (not built).

1.3 Audience

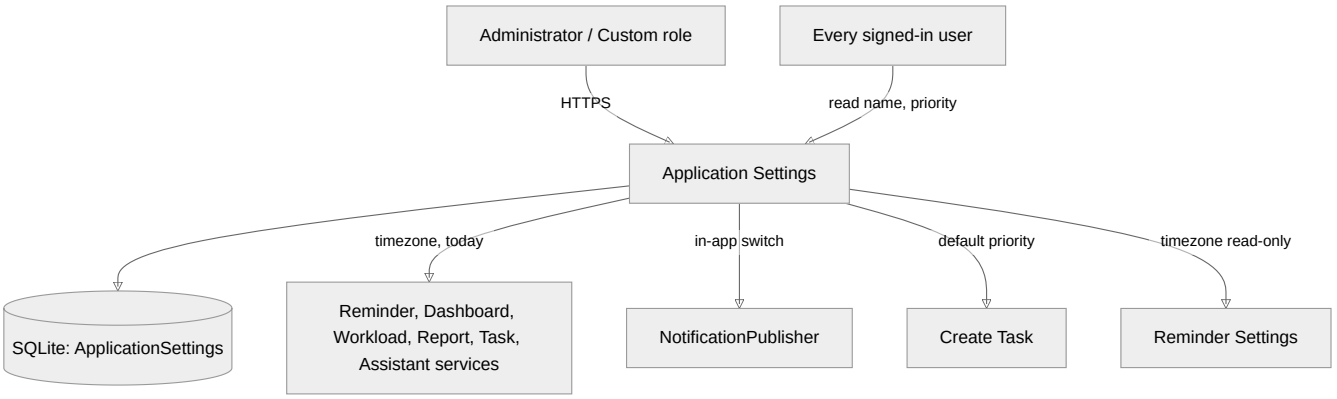
Developer, reviewer and tester of TaskFlow; the product owner for the design decisions.

1.4 Definitions & Abbreviations

Term	Meaning
SPA	Single Page Application (the Angular frontend)
Application clock	`AppClock` (`IAppClock`): resolves the application timezone and the local "today" for all services
Brand name	The application name shown in the sidebar and header
In-app switch	`Notification.InAppEnabled`: whether new bell notifications are created for events
Shell	The authenticated layout (sidebar, header) around every screen

2. System Context

2.1 Context Diagram



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2.2 Actors & External Systems

Actor / system	Interaction	Protocol
Administrator, Custom role with `settings.application.manage`	View and save the four settings	HTTPS / JSON
Every signed-in user (shell, Create Task)	Read the brand name and default priority	HTTPS / JSON
Server-side services (reminder, dashboard, workload, report, task, mail, assistant)	Ask the application clock for the timezone and today	In-process
Notification publisher	Reads the in-app switch before creating an event notification	In-process
Operating system timezone database	Recognises timezone names such as `Asia/Kolkata`	In-process (.NET `TimeZoneInfo`)
External systems	None	–

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3. Architecture Goals, Constraints & Principles

3.1 Quality Goals

Goal	How the architecture supports it
One timezone everywhere	Every service asks the one application clock instead of using server local time (BR-APP-02)
Immediate effect	Values are read from the database on use; no restart or cache invalidation needed (NFR-APP-05)
Reliability	Defaults for missing or unreadable values; an unknown timezone at run time falls back to UTC and is logged (NFR-APP-04)
Security	Read is open to signed-in users but carries no secret; save needs `settings.application.manage` (NFR-APP-02, NFR-APP-03)
Usability	The new brand name appears in the saving user's shell at once (cached signal in the SPA)

3.2 Constraints

- One organization per installation, so one set of application settings.
- Timezone names must be known to the server's operating system timezone database.
- Single API process with one SQLite file; no cache layer.

3.3 Architecture Principles

- Controller → Service → Repository with commits through the unit of work (overall HLD).
- Configuration is data: one row per key in the shared key/value table.
- Time is a dependency: services receive "now" and "today" from `IAppClock`, which also makes them testable.
- The API re-checks every value; the screen's checks are for usability only.

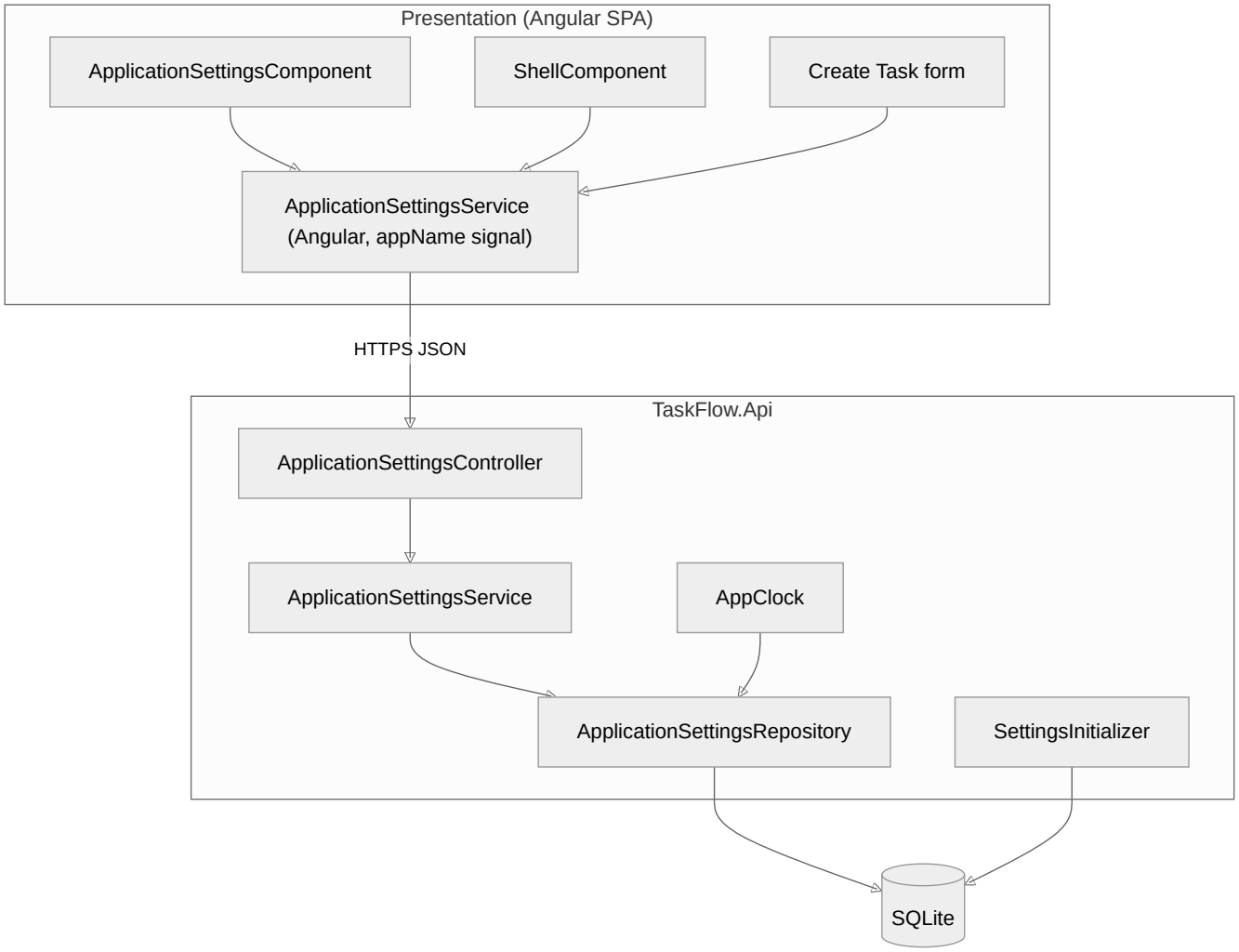
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4. Architecture Overview

4.1 Architecture Style

A screen of the Angular SPA and the shell call one REST controller; the controller delegates to a settings service over the shared key/value table. The application clock is a separate server-side component reading the timezone row and serving every other module in process.

4.2 Logical Architecture



4.3 Layers & Responsibilities

Layer	Responsibility in this sub-module	May call
Presentation	Screen with name, timezone suggestions, priority, in-app switch and link to Reminder Settings; shell brand name; Create Task default priority	API
API / Controllers	`ApplicationSettingsController` : read for any signed-in user, save with `settings.application.manage` , resolves the editor	Services
Validation	Request validator: name 1–100, timezone present and recognised, priority valid, all values present	–

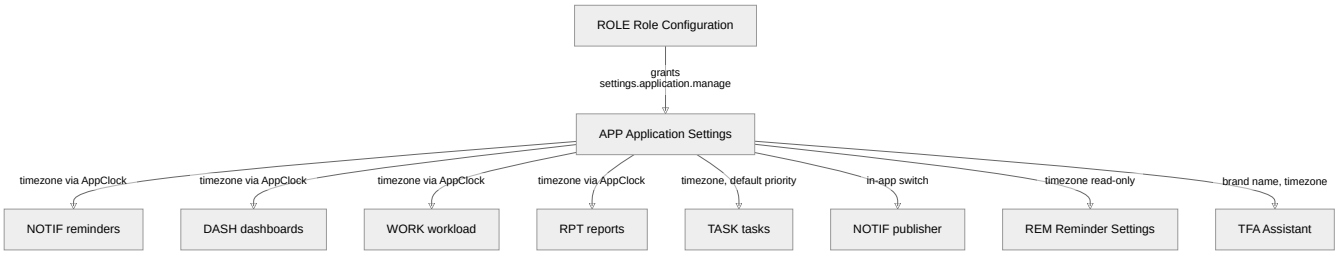
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Layer	Responsibility in this sub-module	May call
Services	`ApplicationSettingsService`: typed read with defaults, full save with editor and time; `AppClock`: timezone and local today with UTC fallback	Repositories
Repositories	`ApplicationSettingsRepository` (shared with Reminder Settings)	Database
Start-up	`SettingsInitializer`: inserts missing rows; the timezone row is seeded from `Application:Timezone`	Database

5. Component / Module Design

Module / component	Responsibility	Key components (high level)	Depends on
Application Settings screen	Show and edit the four values	`ApplicationSettingsComponent`, Angular `ApplicationSettingsService`	Application Settings API
Shell brand name	Show the saved name in sidebar and header	`ShellComponent` reading the cached `appName` signal	Angular `ApplicationSettingsService`
Application Settings API	Authorize, validate and route read/save	`ApplicationSettingsController`, request validator	Settings service
Settings service	Typed read with defaults; full save	`ApplicationSettingsService`	Settings repository, setting key constants
Application clock	Timezone resolution, local today, UTC fallback	`AppClock` (`IAppClock`)	Settings repository, `ApplicationOptions`
Settings storage (shared)	Key/value rows	`ApplicationSettingsRepository`, `ApplicationSetting`	SQLite
Defaults at start-up (shared)	Create missing rows, never overwrite	`SettingsInitializer`	SQLite, `Application:Timezone`

5.1 Module Interaction



Interaction	What is exchanged
APP → NOTIF, DASH, WORK, RPT, TASK, MAIL, TFA	Application timezone and local today through `IAppClock`
APP → TASK (Create Task)	Default task priority, read by the SPA

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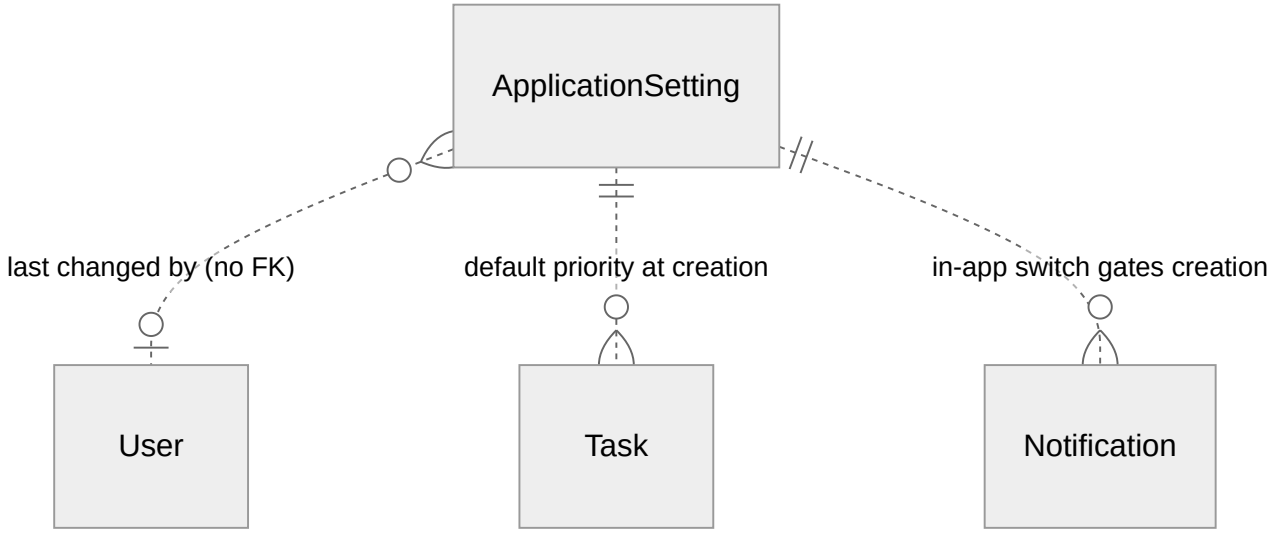
Interaction	What is exchanged
APP → NOTIF (publisher)	In-app switch: off = no new event notification
APP → REM	Timezone shown read-only; reminder intervals are not on this screen (link only)
APP → TFA	Brand name used by the assistant
ROLE → APP	Who may save

6. Data Architecture (high level)

6.1 Data Stores

Store	Technology	Holds	Notes
Main database	SQLite (`taskflow.db`, EF Core)	Four rows in the shared key/value table (`Application.Name`, `Application.Timezone`, `Application.DefaultPriority`, `Notification.InAppEnabled`) with last editor and time	Shared with Reminder Settings
Configuration	`appsettings.json`	`Application:Timezone`: initial and fallback timezone	Not edited at run time
Code defaults	Setting key constants	"TaskFlow", Medium, in-app on	Used for missing or unreadable rows

6.2 Main Data Domains



Domain	Owner	Use here
`ApplicationSetting` (`Application.*`, `Notification.InAppEnabled`)	APP (table shared with REM)	Read and written
`Task` priority	TASK	Not written; Create Task pre-selects the default

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Domain	Owner	Use here
`Notification`	NOTIF	Not written; the publisher checks the in-app switch

6.3 Data Flow

1. Start-up inserts missing rows; the timezone row takes `Application:Timezone`.
2. The shell reads the settings when it mounts and caches the brand name; Create Task reads the default priority.
3. The Administrator saves all four values together; the saving user's shell shows the new name at once.
4. Every "today" decision on the server asks `AppClock`, which reads the timezone row; a change applies on the next call.
5. The notification publisher reads the in-app switch before it writes an event notification.

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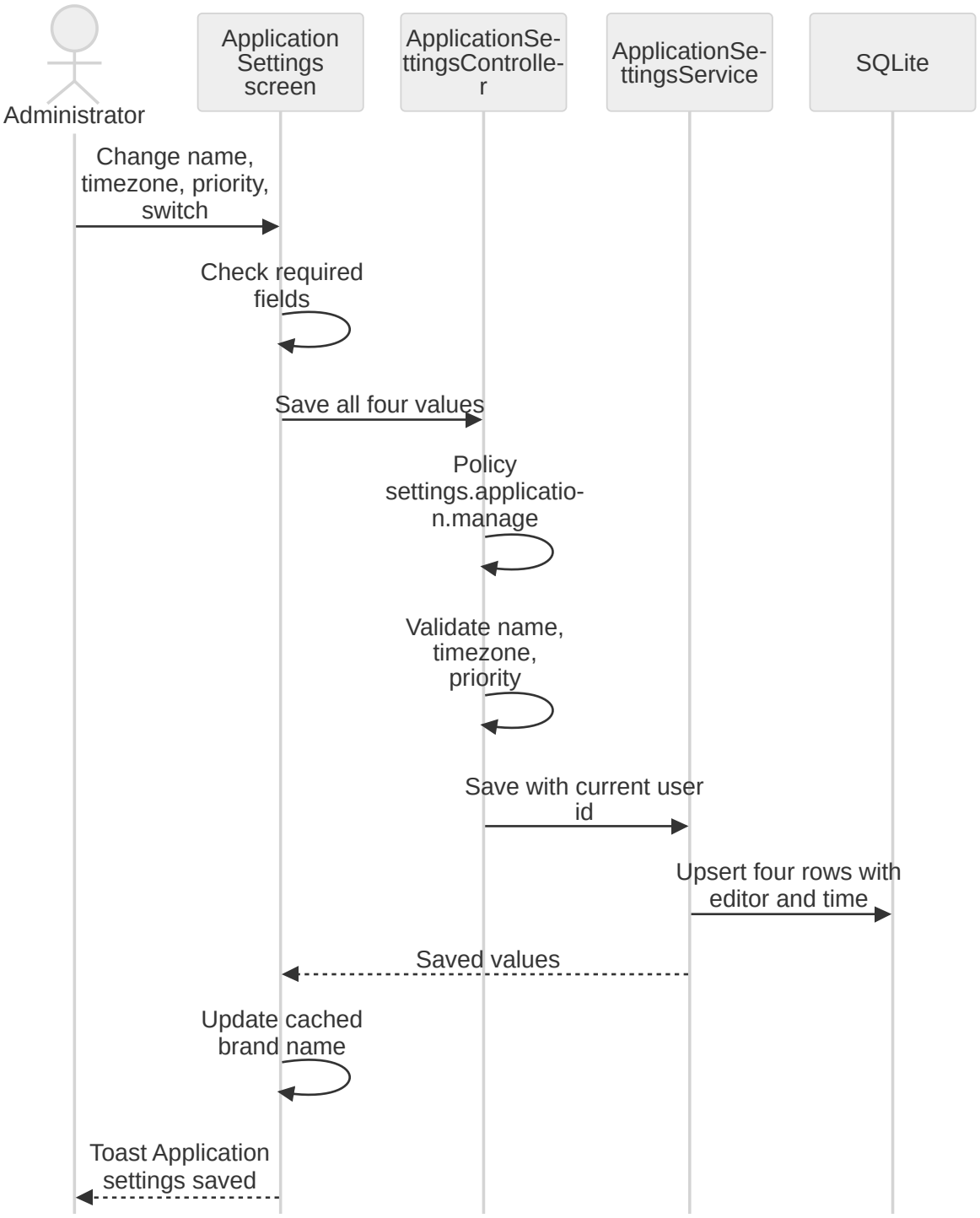
7. Integration Design

Integration	Direction	Purpose	Failure behaviour
.NET timezone database	Outbound (in process)	Validate the timezone on save; resolve it at run time	Unknown on save → 400 "Unknown timezone."; unknown at run time → UTC and an error log entry (FR-APP-09)
Consumers of `IAppClock`	Inbound (in process)	Local today and timezone	Always get a valid timezone (fallback)
Notification publisher	Inbound (in process)	In-app switch	Missing or unreadable value = enabled
Shell and Create Task (SPA)	Inbound HTTPS	Brand name, default priority	Shell keeps "TaskFlow" when the read fails, never blocks
External services	–	None	–

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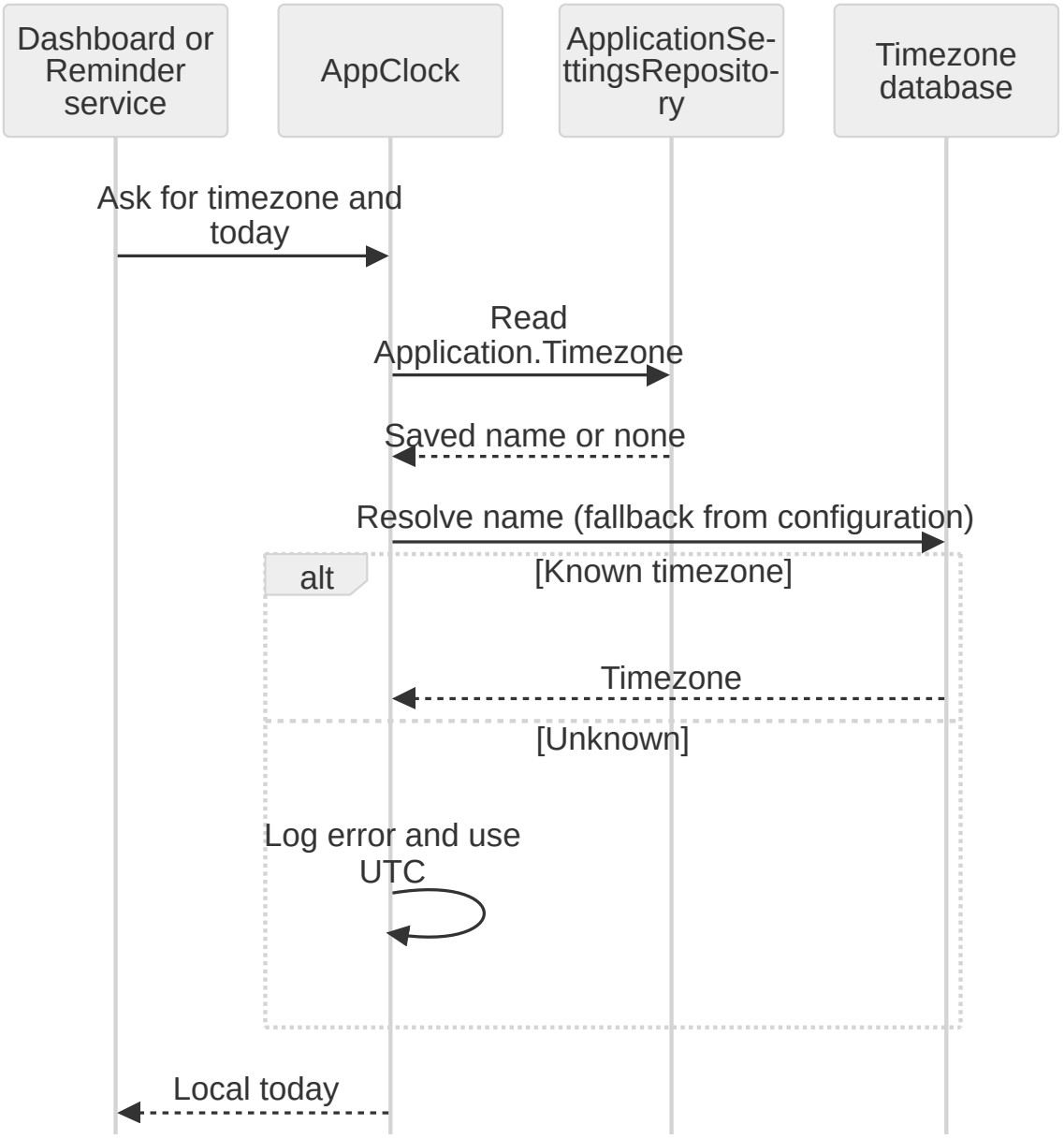
8. Key Flows (high level)

8.1 Save application settings



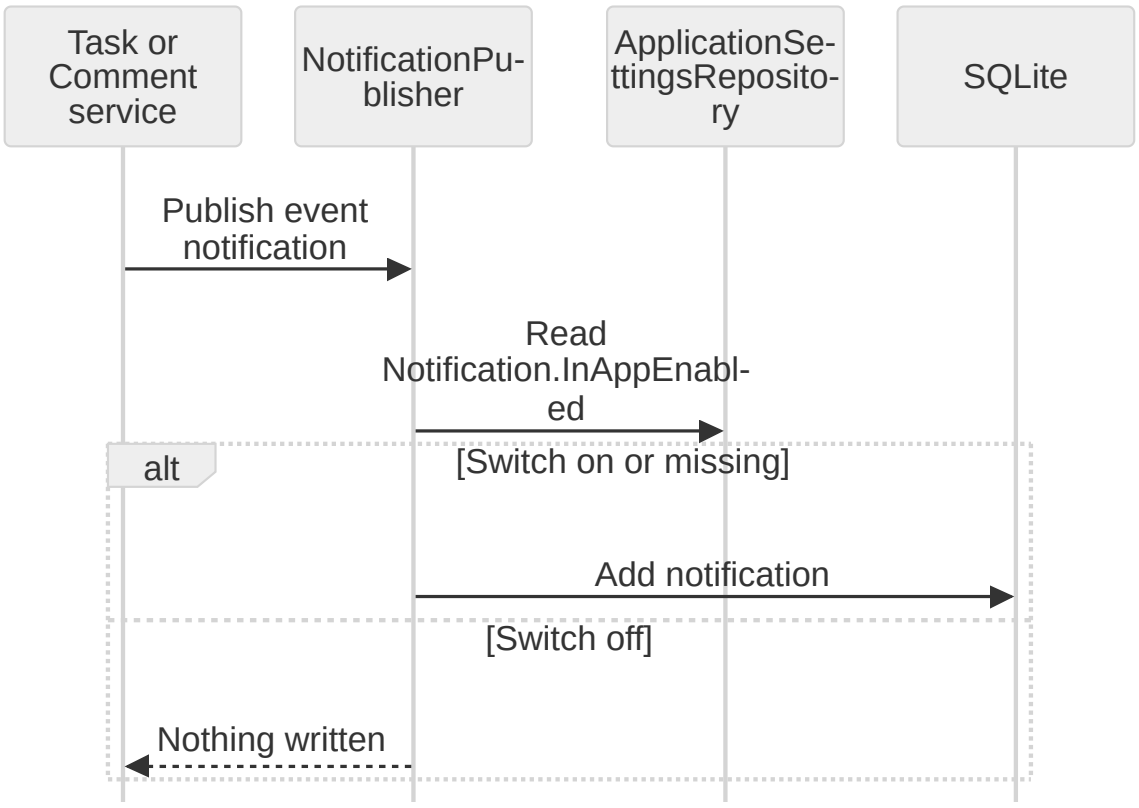
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8.2 A service asks for "today" (application clock)



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8.3 Event notification with the in-app switch



9. Security Architecture

Concern	Approach
Authentication	JWT bearer access token and refresh cookie; session revalidated on every request (overall HLD)
Authorization	Read: any signed-in user (`[Authorize]`), because the shell and Create Task need the name and priority (FR-APP-18). Save: `settings.application.manage` policy (FR-APP-23). Menu item and `permissionGuard` hide the screen from others (FR-APP-22)
Default grants	Administrator only; Manager and Employee denied; custom roles configurable
Data exposure	The readable response contains only the four non-secret values (FR-APP-19, NFR-APP-02)
Input validation	Request validator on save; refusals return 400 with an `errors` map shown next to the fields
Auditing	Last editor and time per row; denied saves logged by the denied-action audit handler; no change history

10. Technology Stack

Application Settings uses the shared TaskFlow stack without additions: Angular 19 SPA (signals for the cached brand name), ASP.NET Core Web API on .NET 10, EF Core 10 with SQLite, FluentValidation 12, and .NET `TimeZoneInfo` for timezone names. Versions and rationale are in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md` , §10).

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11. Deployment Architecture

Application Settings runs inside the single `TaskFlow.Api` process; its values live in the main SQLite database in the data folder. The deployment-specific setting is `Application:Timezone` in `appsettings.json` (shipped as `Asia/Kolkata`; code default `UTC`), which seeds the timezone row on a new installation and is the fallback while the row is missing. The host's timezone database must know the configured names. Environments, hosting and backup are in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`, §11).

12. Non-Functional Design

Area	Design decision
Performance	Read = one query by unique keys; save = one transaction for four rows (NFR-APP-01)
Scalability	Single organization, single instance; a handful of rows
Reliability	Code defaults for missing or unreadable values; UTC fallback for an unknown timezone at run time (NFR-APP-04)
Availability	Shell never waits on the settings call; it keeps "TaskFlow" on failure
Maintainability	All time-of-day logic depends on `IAppClock`; the settings service is the single reader/writer of these keys
Usability	New name visible at once for the saving user; other users see it when their shell next loads (NFR-APP-05)
Observability	Request log per call; error log entry for an unrecognised timezone; denied saves logged
Backup & Recovery	Part of the database file; see overall HLD

13. Design Decisions

ID	Decision	Alternatives considered	Rationale
D-APP-1	Store the four values as rows of the shared key/value table	Typed settings table; `appsettings.json` only	Runtime-editable without migration or restart; shared repository and initializer with Reminder Settings
D-APP-2	One application clock (`IAppClock`) used by every service for "today"	Server local time; per-user timezones	One consistent calendar day for reminders, dashboards, workload, reports and tasks (BR-APP-02); testable time
D-APP-3	Unknown timezone at run time falls back to UTC and is logged	Fail the request; refuse to start	Reminders and dashboards keep working (FR-APP-09)
D-APP-4	Read endpoint open to every signed-in user, save permission-protected	Separate public endpoint for name and priority; permission on read	The shell and Create Task need the values; the response holds no secret

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ID	Decision	Alternatives considered	Rationale
D-APP-5	Cache the brand name in an SPA signal updated on read and save	Reload the page; poll	New name visible immediately for the saving user without a reload
D-APP-6	Save requires all four values	Partial update with defaults	No silent reset of unsent values (BR-APP-03)
D-APP-7	Reminder intervals stay on Reminder Settings; this screen links there	Duplicate them here (original Screen 20)	One owner per setting
D-APP-8	In-app switch checked by the notification publisher, not by each module	Check in every service	One place decides whether an event notification is written

14. Risks, Assumptions & Open Issues

Type	Description	Mitigation / owner
Risk	Wrong timezone saved, so tasks appear due a day early or late (R-APP-01)	Only recognised names accepted; hint on the field – Jojo George
Risk	In-app notifications switched off by mistake (R-APP-02)	Hint on the switch; restricted permission – Jojo George
Risk	No history of earlier values (R-APP-03)	Last editor and time per row – Jojo George
Assumption	The host knows standard timezone names (`UTC`, `Asia/Kolkata`, …)	Jojo George
Open issue	⚠️ To be confirmed: as built, deadline **reminder** notifications are written by the reminder engine directly and do not pass the in-app switch; only event notifications through the publisher are suppressed. FR-APP-13 says "any event".	Jojo George
Open issue	⚠️ To be confirmed: other signed-in users see a new brand name when their shell next loads (sign-in or page reload), not instantly; FR-APP-05 wording.	Jojo George
Open issue	⚠️ To be confirmed: whether a full change history (original Screen 20 FR-6) is still required.	Jojo George
Open issue	Code comments still say "only an Administrator" may save; the code checks `settings.application.manage`.	Jojo George

15. Traceability (requirements → components)

FRD item	Module / component	Section
FR-APP-01	`ApplicationSettingsComponent` → `ApplicationSettingsController` (read) → `ApplicationSettingsService`	§4.2

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FRD item	Module / component	Section
FR-APP-02	`ApplicationSettingsComponent` (loading and error states)	§4.3
FR-APP-03	`ApplicationSettingsComponent`, `ApplicationSettingsService` (save)	§8.1
FR-APP-04	Screen check, request validator, trimmed save in `ApplicationSettingsService`	§4.3
FR-APP-05	Angular `ApplicationSettingsService` cached `appName` signal, `ShellComponent`	§8.1, D-APP-5
FR-APP-06	`ApplicationSettingsComponent` (suggestions); validator timezone check	§7
FR-APP-07	Screen check and request validator	§7
FR-APP-08	`AppClock` used by all services	§8.2, D-APP-2
FR-APP-09	`AppClock` UTC fallback and error log	§8.2, D-APP-3
FR-APP-10	`ApplicationSettingsComponent` (priority list)	§5
FR-APP-11	Create Task form reading the settings	§5.1
FR-APP-12	`ApplicationSettingsComponent`, `ApplicationSettingsService`	§5
FR-APP-13	`NotificationPublisher` checking the in-app switch	§8.3, D-APP-8, §14
FR-APP-14	`ApplicationSettingsComponent` (link to Reminder Settings)	D-APP-7
FR-APP-15	Request validator via the validation filter	§4.3
FR-APP-16	Request validator; `ApplicationSettingsService` full save	D-APP-6
FR-APP-17	`ApplicationSettingsService` defaults; `SettingsInitializer`	§6.3
FR-APP-18	`ApplicationSettingsController` read with `[Authorize]`	§9, D-APP-4
FR-APP-19	Response limited to four non-secret values	§9
FR-APP-20	`ApplicationSettingsService` (editor and time per row)	§6.1
FR-APP-21	`ApplicationSettingsComponent` (saving state, toast, tick, error)	§8.1
FR-APP-22	Navigation builder and `permissionGuard`	§9

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FRD item	Module / component	Section
FR-APP-23	`settings.application.manage` policy on save	§9

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